



CAMER: Contribution-Aware Multimodal Emotion Recognition

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Abstract

Multimodal emotion recognition (MER) integrates data from speech, facial expressions, and text to enhance emotion prediction, with applications in various human-computer interaction scenarios. While recent end-to-end models improved performance, they typically lack interpretability regarding the role of each modality in predicting emotions. This paper introduces a contribution-aware MER (CAMER) with a novel adaptive weighting mechanism that dynamically adjusts the contribution of each modality based on the emotional content using cross-attention between modality features and emotion embeddings. This approach not only enhances the performance of MER by optimizing modality integration but also significantly improves the interpretability of the model's predictions. Evaluations on the IEMOCAP and CMU-MOSEI datasets demonstrate the superiority of our method. Additionally, we provide an interactive demo that allows users to explore and visualize the model's decision-making process.

Index Terms: multimodal emotion recognition, interpretability, human-computer interaction

1. Introduction

Multimodal emotion recognition (MER) has gained significant attention in recent years due to its potential in enhancing human-robot interaction [1], personalized services [2], and mental health monitoring [3]. By integrating information from multiple modalities such as speech [4], facial expressions [5], and textual data [6], MER models can capture a more comprehensive understanding of human emotions.

Historically, most multimodal emotion recognition (MER) approaches followed a two-stage pipeline, where features were first extracted separately for each modality, followed by a fusion stage to predict the final emotion. Early feature extraction relied on hand-crafted algorithms, such as OpenFace for facial features [7], COVAREP for acoustic analysis [8], and GloVe for textual representations [9]. Initially, MER methods employed simple fusion strategies, including early fusion—where features were concatenated before classification [10]—and late fusion, which combined modality-specific predictions [11]. However, more sophisticated approaches soon emerged. For example, tensor fusion network [12] performed Cartesian product for tri-modal dynamics, while multimodal transformer (MulT) [13] handled unalignment in multimodal data with cross-modal attention. [14] introduced multimodal adaptation gate with pre-trained language models. Also, EmoEmbs [15] utilized the relationship between different emotion categories by exploiting modal-wise emotion embeddings.

Although this pipeline-based approach showed promise, recent advances in end-to-end learning methods have significantly

improved performance. These methods jointly trained feature extraction and fusion mechanisms, allowing the network to learn modality interactions in a more integrated and effective manner. [16, 17] presented end-to-end audio-visual MER models. FE2E [18] first proposed an end-to-end MER model that utilizes the three major modalities: visual, acoustic, and textual. V2EM [19] introduced hierarchical attention to enhance performance in the acoustic modality. MLMER [20] employed a transformer encoder to encode multimodal features simultaneously and allowed label-level representations to query these features through a transformer decoder. VMER [21] exploited modal-specific large pre-trained models to significantly boost the performance.

However, despite the advancements in prediction performance, most of these works focus solely on predicting the final emotion label. This focus often leaves it unclear which modality contributes the most to the overall prediction, making it difficult to interpret the model's decisions or understand the role of each modality in the emotion recognition process. Moreover, while research in multimodal emotion recognition has made significant strides, there remain notable discrepancies in translating these advancements into real-world applications. Many existing models are developed and evaluated in controlled experimental settings, limiting their practical usability in interactive systems.

To address this gap, this paper proposes a novel approach that leverages adaptive weights between emotion embeddings and modality-specific representative features. This method directly quantifies the contribution of each modality to the prediction of each emotion, offering a more interpretable model for emotion recognition. By using cross-attention mechanisms, we dynamically adjust the importance of each modality depending on the emotional content, providing deeper insights into how specific modalities influence the prediction process. Further, an interactive demo is provided, allowing users to visualize the model's decision-making process. With test on real-world YouTube downloaded videos, the proposed demo serves as a baseline for the development of deployable MER systems, connecting research advancements into real-world applications.

- We introduce an adaptive weighting mechanism for end-to-end MER that dynamically weighs the contribution of each modality, offering interpretability and enhanced performance.
- We perform experiments on benchmark datasets such as IEMOCAP and CMU-MOSEI to validate the effectiveness of our approach and demonstrate its superiority over traditional methods.
- We provide an interactive demo framework, allowing users to visualize and interact with our cross-attention-based model, making it applicable to real-world applications.

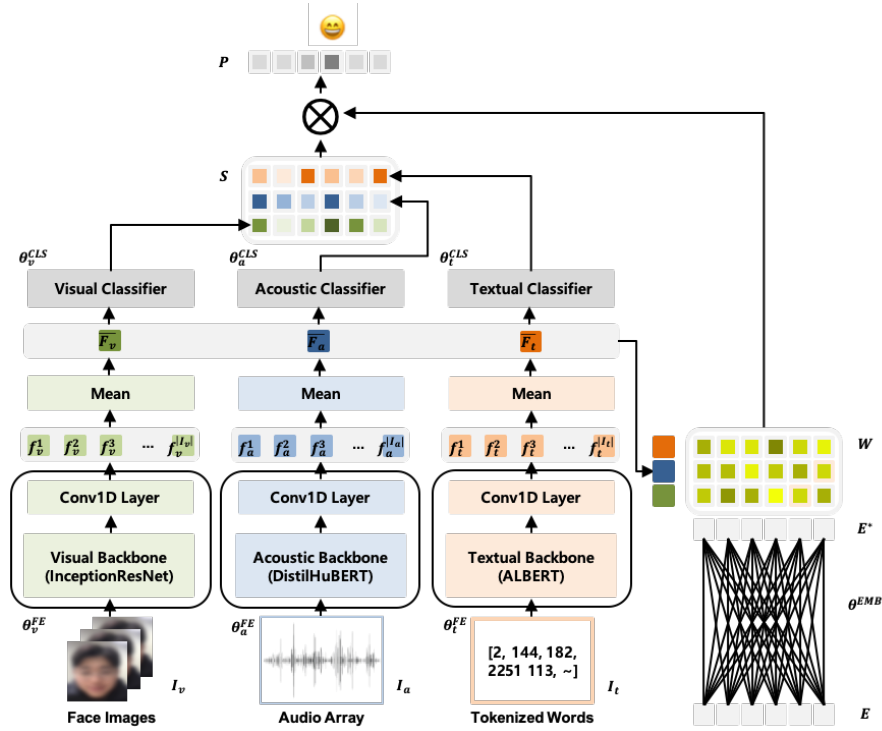


Figure 1: Overall structure of CAMER

2. Methods

2.1. Problem Formulation

Given the sequences of multimodal inputs $I_{m \in \{v,a,t\}}$, the problem of end-to-end MER can be formulated as making the prediction $P \in \mathbb{R}^{|c|}$, where m indicates one of the modalities among visual, acoustic and textual, and c implies emotion classes. In order to minimize the human intervention for input pre-processing, cropped face images, raw audio input, and tokenized texts are used as I_v, I_a, I_t , respectively.

2.2. Modal-wise Feature Extraction

For each modality, given I_m , the modal-wise features F_m are extracted as follows:

$$F_m = \theta_m^{FE}(I_m) = \{f_m^1, f_m^2, \dots, f_m^{|I_m|}\} \in \mathbb{R}^{|I_m| \times d}$$

where θ_m^{FE} is composed of a modal-wise backbone and convolution 1D layer to map the multimodal features to the same embedding dimension d . Following [21], strong pre-trained models for each modality are utilized as backbone models to maximize the model performance. For the visual modality, InceptionResNet [22] trained on facial recognition data called VggFace2 is used. For the acoustic modality, DistilHubert [23] is used, which can take raw audio as an input. For the text modality, ALBERT [24] is used. Given F_m , the modal-representative feature is driven as follows:

$$\bar{F}_m = \frac{1}{|I_m|} \times \sum_{i=1}^{|I_m|} f_m^i$$

2.3. Emotion Embedding Vectors

Inspired by previous works [25, 20], emotion embedding vectors $E \in \mathbb{R}^{|c| \times d}$ are first initialized. Then, in order to learn

the correlation among the emotions, the embedding vectors are refined using basic transformer encoder layers [26] as $E^* = \theta^{EMB}(E)$.

2.4. Emotion-Modal adaptive weight

The emotion-modal adaptive weight $W \in \mathbb{R}^{|c| \times |m|}$ can be driven by the scaled dot-product attention [26]:

$$W = \text{softmax}\left(\frac{(W_Q E^*)(W_K [\bar{F}_v; \bar{F}_a; \bar{F}_t])^T}{\sqrt{d}}\right)$$

where the emotion embedding vectors and concatenated modal-representative features are considered as the query and key in cross-modal attention with W_Q and W_K the weight matrix for query and key. The adaptive weight W has a property that the sum of weight values over each emotion equals to 1, implying that each weight can successively indicate the relative importance of each modality to each emotion.

2.5. Final Prediction

For the final prediction, the modal-wise classifier first outputs modal-wise prediction score:

$$S_m = \theta_m^{CLS}(\bar{F}_m) \in \mathbb{R}^{|c| \times 1}$$

After stacking the modal-wise prediction scores $S \in \mathbb{R}^{|c| \times |m|}$, element-wise multiplication is applied to drive the weighted scores $S^* = W \circ S$. The weighted scores directly reflect the contribution of each modality to the final prediction $P = \sigma(\sum_{i=1}^{|m|} S_{*,i}^*)$.

3. Experiments

3.1. Experimental Settings

Following the previous works [18], the restructured IEMOCAP and CMU-MOSEI datasets were used for the experiments. The reorganized IEMOCAP dataset contains 7,380 data samples (5,162 for training, 737 for validation and 1,481 for testing) in total, which are labeled in six categories of anger, excitement, frustration, happiness, neutral and sadness. And the reorganized CMU-MOSEI dataset is composed of 20,477 data samples (14,524 for training, 1,765 for validation and 4,188 for testing), which are labeled in six categories of anger, disgust, fear, happiness, sadness and surprise. Due to the class imbalance, F1 score was adopted for evaluation. As inputs, CAMER took 100 tokens of words, 5 face images of size 160×160 and maximum 10 seconds of audio array with a sampling rate of 16 kHz. These inputs were either zero padded or truncated depending on their original length. The loss function was a binary cross-entropy loss, where the weighting [18] was applied to mitigate the data imbalance. The experiments were conducted on one Nvidia RTX 3090 GPU with the hyperparameters as following: batch size of 8, learning rate of $5e-6$ for the backbone models and $5e-5$ for the rest parts with Adam optimizer [27] and epoch of 10.

3.2. Results and Analysis

3.2.1. Quantitative Results

The experimental results on the IEMOCAP and CMU-MOSEI datasets are summarized in Table 1 and 2, respectively. One thing to note is that adopting large pre-trained models significantly increase the performance as can be observed in VMER and CAMER. Also, CAMER shows improved performance which might be due to the exploitation of the correlation between modal representative features and emotion embedding vectors through adaptive weighting.

3.2.2. Qualitative Results

The qualitative results of CAMER on the CMU-MOSEI test samples are shown in Figure 2. For each sample, the scores S of each modality before applying the weights are shown above, and the scores S^* of each modality after applying the weights W are shown below. Based on the CAMER model design, it can be seen that predictions were first made according to the context of each modality. For example, in Figure 2a, anger and disgust were prominently observed in the textual modality, while unhappiness was prominent in the visual and acoustic modalities. The final scores, reflecting the weights, indicate that the textual modality played a major role in the final prediction results for anger and disgust, while the visual and acoustic modalities significantly influenced sadness. Similar examples can be seen in Figure 2b, where the visual feature played a significant role in the emotion of happiness.

3.2.3. Emotion Embeddings

Next, an analysis was conducted to examine the structure of the trained emotion embedding vectors E^* . To facilitate an intuitive comparison between them were measured. The results are shown in Figure 3. In the case of IEMOCAP dataset, it was observed that the emotions typically perceived as close, such as anger and frustration, became closer after training, while anger and happiness maintained the distance. Additionally, sadness was moving

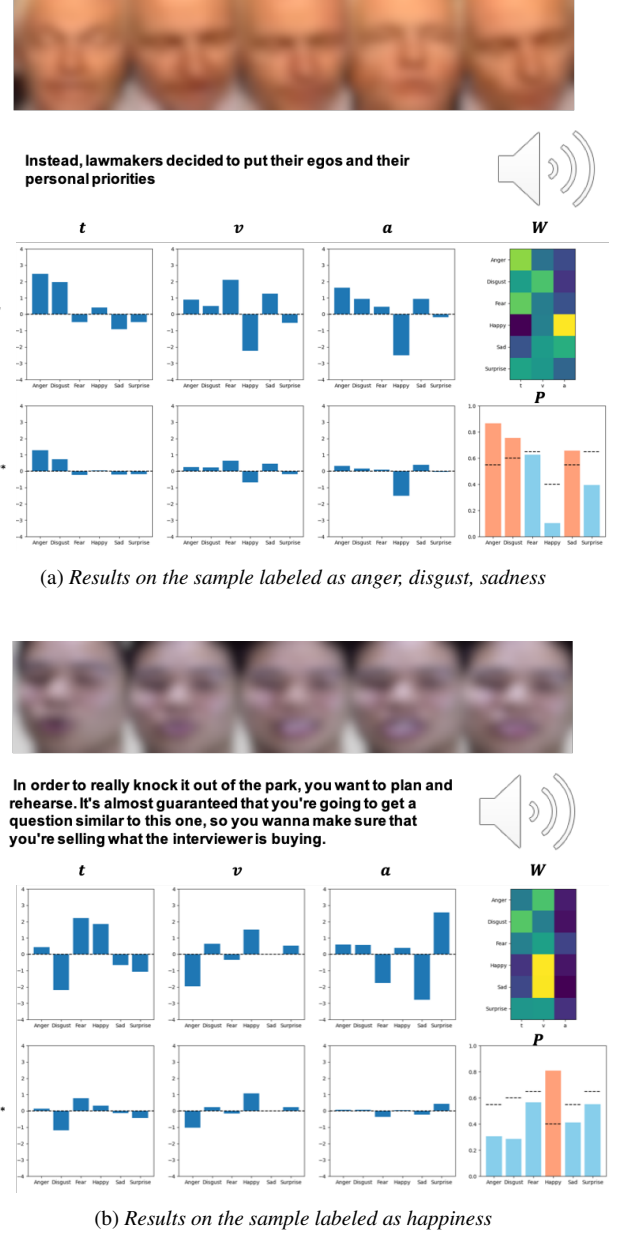


Figure 2: Qualitative results on CMU-MOSEI

farther from heightened emotional states such as excitement and frustration. In the CMU-MOSEI dataset, the distances between happiness and emotions like anger and disgust increased, while the distance between sadness and fear decreased. These results indicate that the trained emotion embedding vectors exhibit a tendency where negative emotions cluster closer together, while positive emotions maintain a distinct separation from negative ones.

3.2.4. Interactive Demo

Figure 4 shows the interactive demo of CAMER by Graidio¹ with a random sample downloaded from YouTube. A user can drag and drop a video sample to evaluate in the right window

¹<https://www.gradio.app/>

Model	Anger	Excitement	Frustration	Happiness	Neutral	Sadness	Average
EmoEmbs	48.9	58.3	52.0	38.3	48.7	53.0	49.8
MuT	60.7	58.0	57.0	46.8	53.7	65.4	56.9
FE2E	61.5	60.7	58.6	44.7	55.8	68.1	58.2
V2EM	60.6	57.0	58.8	45.4	54.2	59.4	55.9
MLMER	66.8	66.8	57.0	48.5	56.6	69.6	60.9
VMER	66.1	66.4	62.5	47.7	64.5	74.0	63.5
CAMER	68.1	69.6	62.1	47.6	61.2	75.1	63.9

Table 1: *Quantitative results on IEMOCAP*

Model	Anger	Disgust	Fear	Happiness	Sadness	Surprise	Average
EmoEmbs	49.4	48.7	23.4	71.9	47.5	24.0	44.2
MuT	47.5	49.3	25.3	75.4	48.3	25.6	45.2
FE2E	50.0	56.6	26.1	72.6	49.0	26.7	46.8
V2EM	49.3	56.4	27.7	72.4	48.2	27.2	46.8
MLMER	50.2	57.0	29.0	72.6	49.2	27.6	47.6
VMER	52.9	58.0	28.5	75.5	51.5	30.3	49.4
CAMER	53.5	57.4	32.3	75.6	51.3	28.6	49.8

Table 2: *Quantitative results on CMU-MOSEI*

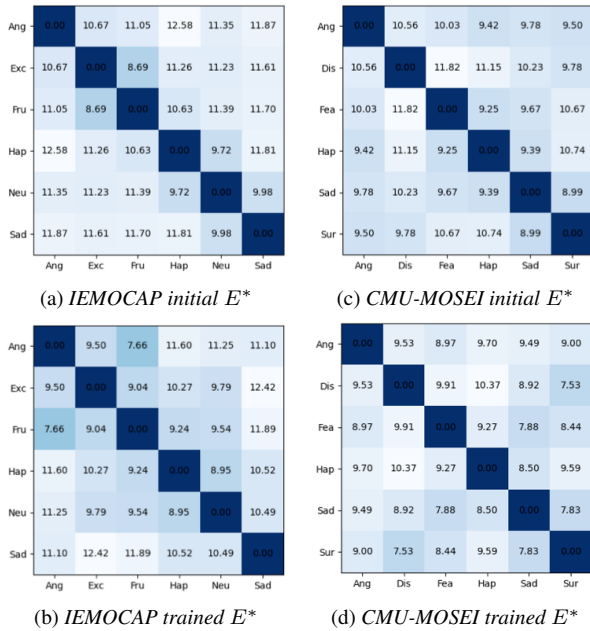


Figure 3: *Euclidean distances between learned emotion embedding vectors*

of the system. Given the raw video input, the system pre-processes each visual, acoustic and textual modality through face detection, voice activity detection and speech recognition. For each model, MTCNN [28], Silero-VAD² and Whisper [29] are adopted. The prediction results are displayed in the right windows with modal-wise and overall predictions.

4. Conclusion

In this paper, we proposed a novel adaptive weighting mechanism for multimodal emotion recognition (MER), which dy-

²<https://github.com/snakers4/silero-vad>

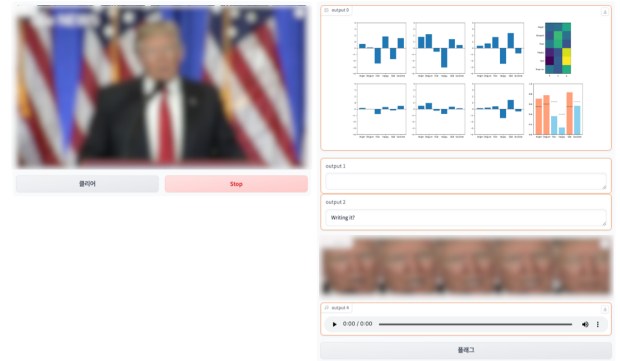


Figure 4: *Interactive demo of CAMER*

namically adjusts the contribution of each modality using cross-attention between modality-specific features and emotion embeddings. CAMER directly addresses the limitations of traditional fusion methods by providing both improved performance and interpretability. Through extensive evaluation on benchmark datasets like IEMOCAP and CMU-MOSEI, we demonstrated the superiority of our model. Additionally, we introduced an interactive demo framework, offering practical insights into how the model makes emotion predictions, enhancing its real-world applicability. However, despite these advancements, there are several limitations to be addressed. While the datasets used for the experiments are among the largest MER benchmark datasets available, challenges such as an insufficient number of data samples and class imbalance remain. Future work will focus on gathering high-quality, diverse MER data to improve robustness and generalization. Moreover, while CAMER provides modality-level contribution analysis, extending this approach to offer sample-specific interpretability could further enhance the transparency of emotion recognition models. By addressing these challenges and considerations, future work can further improve the reliability and ethical deployment of MER systems, paving the way for more practical and responsible real-world applications.

5. References

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